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EXPERIMENT 5

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Subject Name: Computer Networks

Subject Code: 24CSH-337

1. Aim/Overview of the practical

To configure and understand Dynamic Routing using RIP (Routing Information Protocol) on multiple routers, and to verify inter-network communication in Cisco Packet Tracer.

2. Software/Tools Required

- Cisco Packet Tracer
 - Windows Operating System
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3. Theory

A) What is Routing?

Routing is the mechanism by which a router decides how to forward a packet from the network it originated on toward the network where the destination host resides. Whenever two hosts sit on different subnets, an intermediate router must consult its routing table to work out the correct next-hop for the packet.

B) What is Dynamic Routing?

Dynamic Routing lets routers build and maintain their routing tables on their own, by periodically exchanging information about the networks they know about with directly connected neighbors. This removes the need for an administrator to type in every individual route by hand, as would be required under static routing.

Benefits of Dynamic Routing

Benefit	What it Means
Self-Learning	Routers discover neighboring networks on their own
Reduced Admin Effort	No need to type a route for every possible network
Resilience	Can reroute traffic automatically if a link fails
Grows Well	Handles large, frequently changing networks better than static routing

Common Dynamic Routing Protocols

Protocol	Stands For
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RIP	Routing Information Protocol
OSPF	Open Shortest Path First
EIGRP	Enhanced Interior Gateway Routing Protocol
BGP	Border Gateway Protocol

C) RIP (Routing Information Protocol)

RIP is a distance-vector protocol and one of the earliest dynamic routing protocols still used today. Each router running RIP periodically broadcasts its routing table to its directly connected neighbors, and those neighbors merge the received routes into their own tables.

Why is RIP still used?

Static routing quickly becomes unmanageable as the number of networks grows. RIP addresses this by letting routers:

- Discover neighboring networks automatically.
- Keep their routing tables current without manual edits.
- Pick a best path automatically based on hop count.

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How RIP Chooses a Path

RIP measures distance in hop count — the number of routers a packet must pass through to reach the destination network. Given a chain such as:

$R1 \rightarrow R2 \rightarrow R3 \rightarrow \text{Network } X$

the hop count from R1's perspective is 2, since the packet passes through R2 and R3 before arriving.

RIP Hop Count Reference

Hop Count	Interpretation
0	Directly connected network
1	Reachable through one router
4	Reachable through four routers
15	Farthest network RIP will still advertise
16	Treated as unreachable (infinite metric)

Sample RIP Routing Table

Destination Network	Next Hop	Hop Count
172.16.1.0	Directly Connected	0
172.16.2.0	10.1.1.2	1
172.16.3.0	10.1.1.2	2

Static Routing vs Dynamic Routing

Feature	Static Routing	Dynamic Routing
Setup	Every route entered by hand	Routes learned automatically once the protocol is enabled
Reacts to Failure	No — needs manual correction	Yes — recalculates around failed links
Best Fit	Small, unchanging networks	Larger or frequently changing networks
Processing Load	Minimal	Ongoing, due to periodic route calculation
Memory Needed	Minimal	Higher, to hold learned routes
Traffic Overhead	None	Periodic routing updates consume some bandwidth
Exposure	Topology stays private	Neighbors learn about the network layout
Upkeep	Gets harder as the network grows	Scales with far less manual effort

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4. Steps for Experiment/Practical

A) Dynamic Routing using 6 PCs, 3 Routers and 3 Switches

Procedure / Steps:

1. Open Cisco Packet Tracer and place 6 PCs, 3 Routers, and 3 Switches on the workspace.
2. Cable all devices together to match the intended topology.
3. Assign IP addresses to every PC and configure IP addresses on the router interfaces.
4. Set the Default Gateway on each PC.
5. Enable RIP on all three routers and advertise each router's directly connected networks.
6. Check the resulting routing table with: show ip route
7. Verify connectivity between PCs on different networks using the ping command.

B) Dynamic Routing using 8 PCs, 4 Routers and 4 Switches

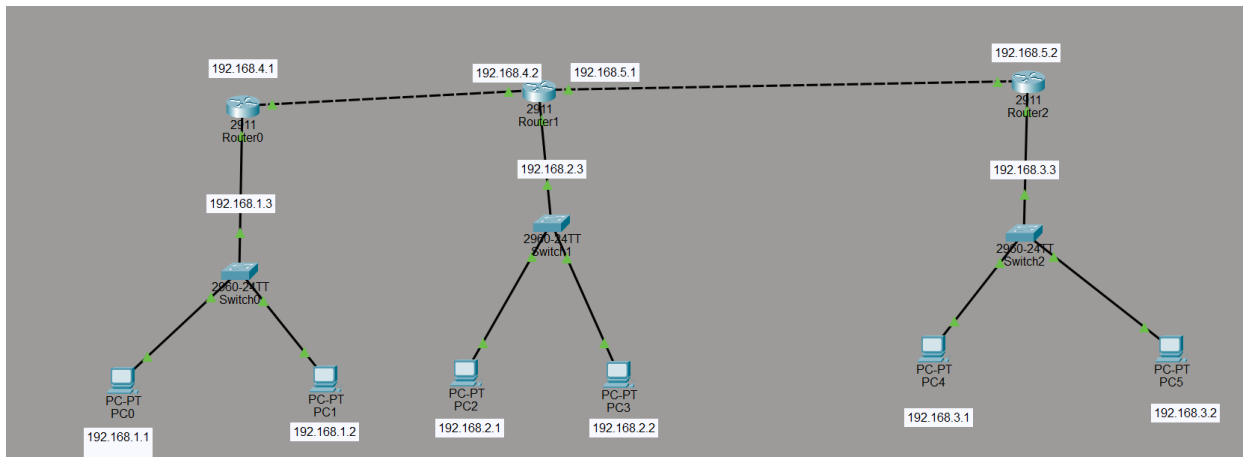
Procedure / Steps:

1. Open Cisco Packet Tracer and place 8 PCs, 4 Routers, and 4 Switches on the workspace.
2. Cable all devices together to match the intended topology.
3. Assign IP addresses to every PC and configure IP addresses on the router interfaces.
4. Set the Default Gateway on each PC.
5. Enable RIP on all four routers and advertise each router's directly connected networks.
6. Check the resulting routing table with: show ip route
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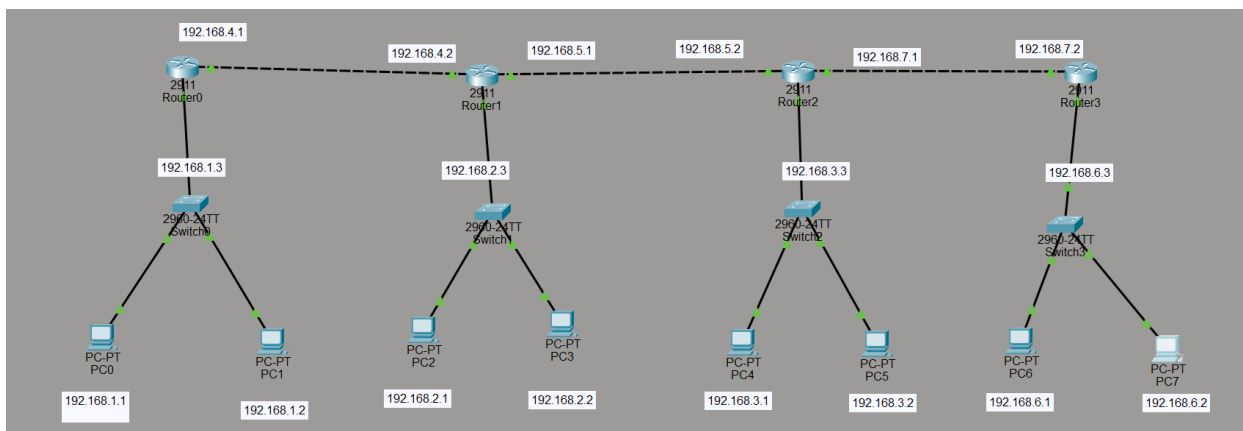
5. Output

Dynamic Routing using 6 PCs, 3 Routers and 3 Switches



Router0, Router1, and Router2 are chained together over 192.168.4.0/24 and 192.168.5.0/24 WAN links. Each router also serves its own LAN through a switch: Switch0 (192.168.1.0/24) with PC0 and PC1, Switch1 (192.168.2.0/24) with PC2 and PC3, and Switch2 (192.168.3.0/24) with PC4 and PC5. With RIP enabled on all three routers, each router advertises its directly connected networks so that every PC can reach every other PC without any manually entered static route.

Dynamic Routing using 8 PCs, 4 Routers and 4 Switches



Router0 through Router3 are chained in series across 192.168.4.0/24, 192.168.5.0/24, and 192.168.7.0/24 WAN links. Each router carries its own LAN: Switch0 (192.168.1.0/24) with PC0/PC1, Switch1 (192.168.2.0/24) with PC2/PC3, Switch2 (192.168.3.0/24) with PC4/PC5, and Switch3 (192.168.6.0/24) with PC6/PC7. Extending RIP to all four routers lets every router automatically learn the routes to the other three LANs, and every PC across the four subnets is able to reach every other PC.

6. Result

Dynamic Routing using RIP (Routing Information Protocol) was implemented and verified successfully in Cisco Packet Tracer across both the three-router and four-router topologies. RIP was enabled on every router, each

router's routing table populated automatically once neighboring networks were advertised, and connectivity between PCs on different subnets was confirmed using the ping command.

7. Learning Outcomes

- Understood the concept and importance of routing in computer networks.
- Learned how Dynamic Routing differs from Static Routing.
- Understood how RIP (Routing Information Protocol) operates as a distance-vector protocol.
- Learned how RIP uses hop count to choose the best path to a network.
- Configured RIP across multiple routers and observed automatic route learning and updates.
- Learned to inspect a router's routing table using the show ip route command.
- Verified end-to-end communication across different networks using the ping command.